Experiment 5: Band Structure Simulation for Graphene and CNTs Using NanoHUB

Aim:

To simulate and study the band structure of nanomaterials such as graphene and Carbon Nanotubes (CNTs) using the NanoHUB online tool.

Software Required:

NanoHUB Online Tool – ABACUS (Assembly of Basic Applications for Coordinated Understanding of Semiconductors).

Theory

Band structure describes the allowed energy levels of electrons in a material as a function of their wave vector. It helps determine whether a material behaves as a conductor, semiconductor, or insulator.

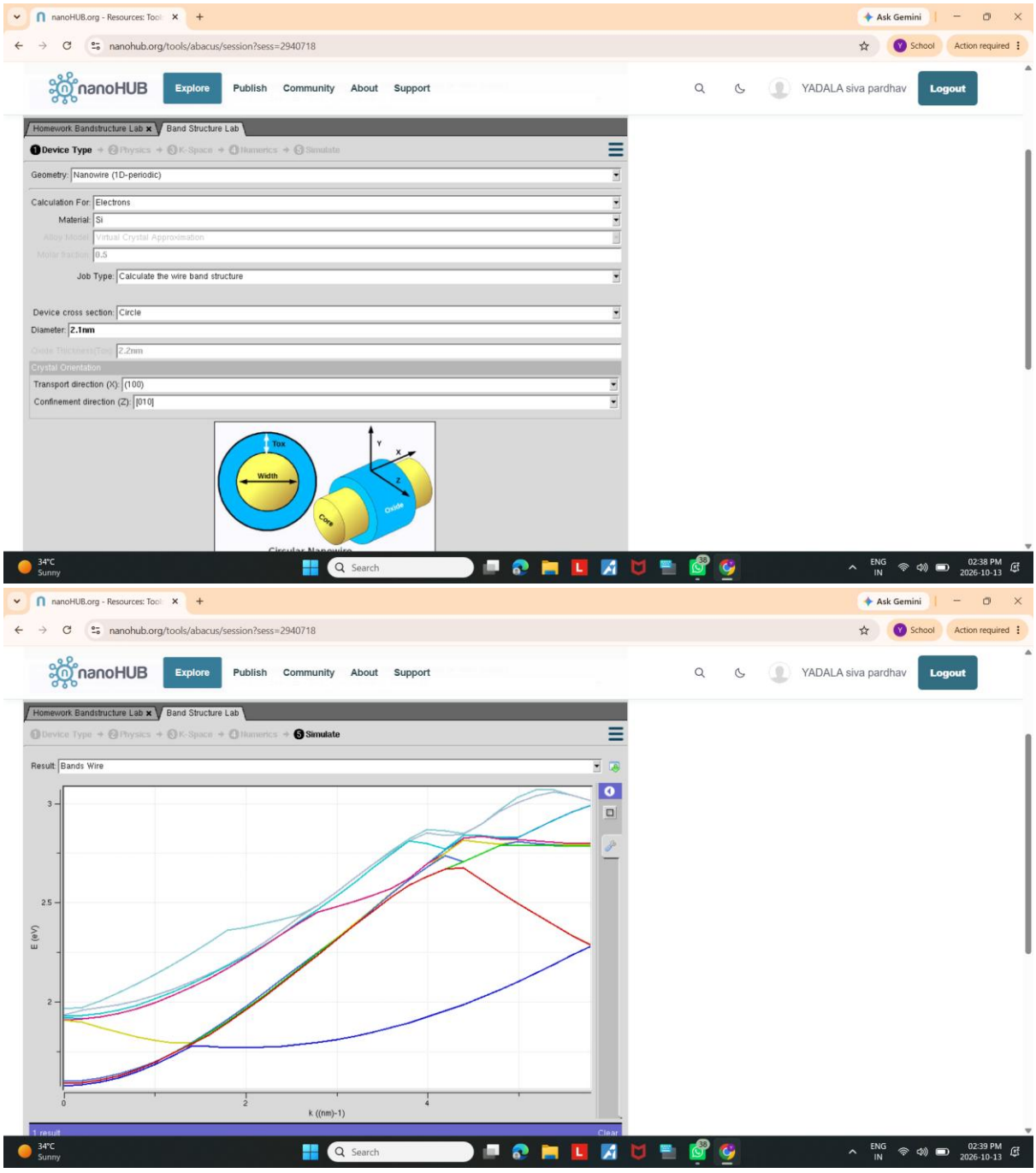
Graphene: Graphene is a single layer of carbon atoms arranged in a hexagonal lattice. It has a zero band gap at the Dirac points, making it a semimetal.

Carbon Nanotubes (CNTs): CNTs are cylindrical structures formed by rolling graphene sheets. Their electronic properties depend on their diameter and chirality. CNTs can behave as either metals or semiconductors.

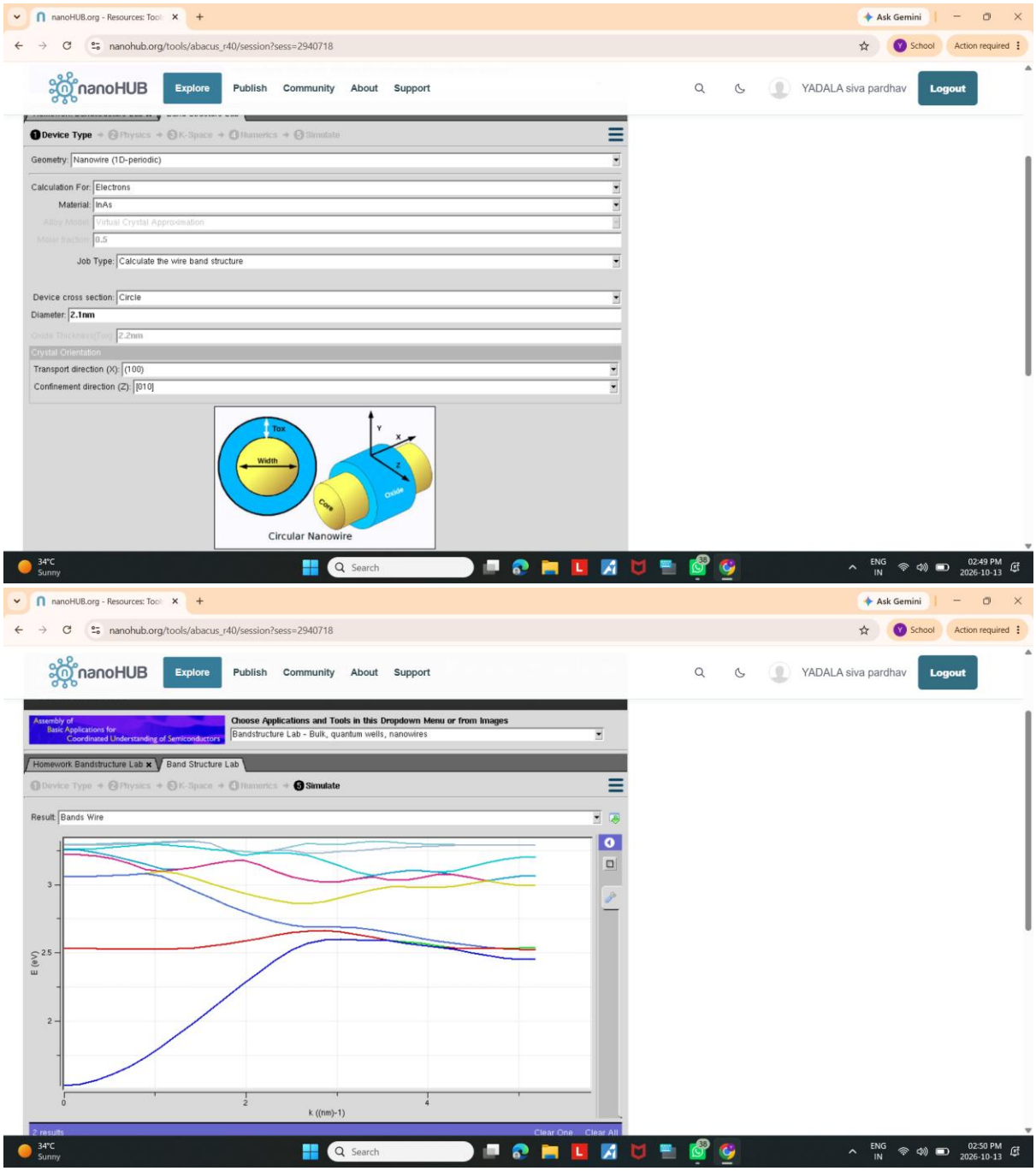
Quantum confinement: When the dimensions of a nanomaterial become very small, electron motion is restricted, resulting in discrete energy levels and changes in its band structure.

Procedure

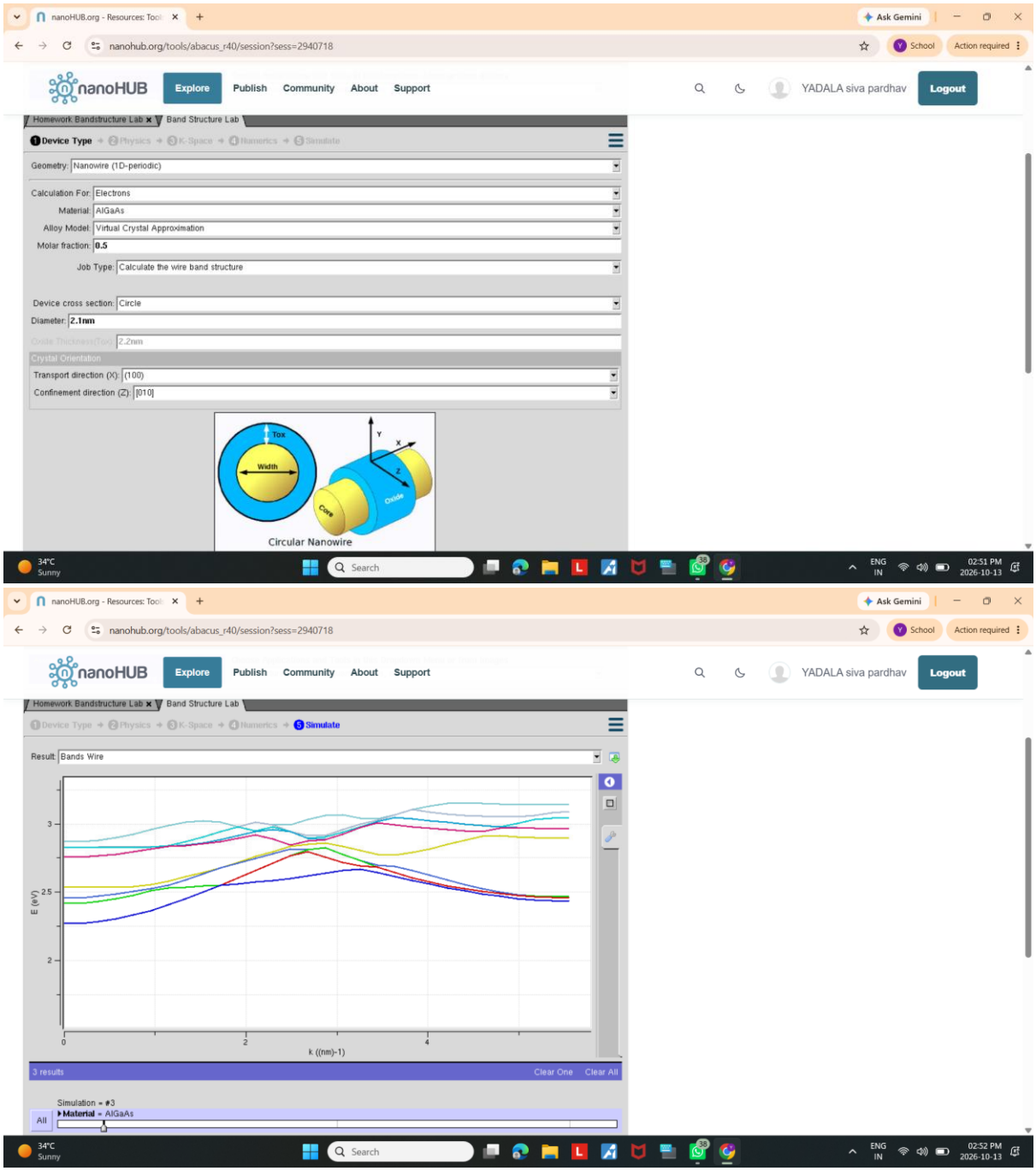
1. Open the NanoHUB website: https://nanohub.org/
2. Navigate to Resources → Tools.
3. Search for the ABACUS tool.
4. Launch the tool from its information page.
5. Select Applications → Band Structure Lab.
6. Choose the appropriate geometry or structure for graphene or CNTs, if available.
7. Select the material and required simulation parameters.
8. Click Simulate to obtain the band structure.
9. Observe the energy bands and band gap.
10. Repeat the simulation for different structures and compare the results.



Case Study 1: Graphene



Case Study 2: Carbon Nanotubes (CNTs)



Note: These are theoretical expectations, not measured simulation results. Actual outputs depend on the chosen structure and simulation settings.

Result

The band structures of graphene and carbon nanotubes were studied using a NanoHUB simulation tool, and their electronic properties were compared based on their band gaps.